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ITMO UNIVERSITY

International Research and Educational Center
for Physics of Nanostructures

LAB 2
for the Course
*"Discrete Control Systems"*

on the topic:
# Second order linear discrete systems dynamic properties study


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# INTRODUCTION

## 0.1 Introduction

The purpose of this work is to study the dynamic properties of second order linear discrete systems through analysis and synthesis of state feedback controllers. The control object under consideration is described by a second-order differential equation, which is converted to discrete form using matrix exponential methods.

The main objectives of this laboratory work are:

- To develop mathematical models of the control object in both continuous and discrete time domains
- To design a linear state feedback controller that places closed-loop poles at specified locations
- To analyze the system behavior for different root locations in the z-plane
- To verify the controller performance through simulation and analysis of transient responses and phase trajectories

The work follows a systematic approach including derivation of state-space models, discretization using matrix exponential methods, synthesis of feedback gains for pole placement, and comprehensive analysis of system dynamics. The MATLAB environment is utilized for numerical computations and system simulations.

## 0.2 Notations

Throughout this work, the following notations are used:

Table 1 — Table 1: Notations of symbols

| Symbol | Explanation |
| --- | --- |
| $x$ | State vector |
| $y$ | Output variable |
| $u$ | Control input |
| $A, B, C$ | State-space matrices (continuous time) |
| $A_d, B_d$ | State-space matrices (discrete time) |
| $T$ | Sampling time |
| $K$ | Linear state feedback matrix |
| $F_d$ | Closed-loop system matrix |
| $z$ | Complex variable in z-domain |

## 0.3 Variant

Table 2 — Table 2: Parameters for Variant 22

| Parameter | Ts / Root Locations ($z_1, z_2$) |
| --- | --- |
| $T$ (s) | 0.25 |
| Exp 1 | 0.94, -0.64 |
| Exp 2 | -1.02, 1.25 |
| Exp 3 | -0.69, 0.01 |
| Exp 4 | ±0.25j |
| Exp 5 | -0.07±0.88j |

# 1 SOLUTIONS

The purpose of the work: To study the dynamic properties of second order linear discrete systems and synthesize state feedback controllers for pole placement.

## 1.1 Methodology

The synthesis methodology includes:

- Development of continuous state-space model for the second-order system
- Discretization using matrix exponential methods
- Synthesis of linear state feedback controllers using pole placement
- Analysis of system behavior for different root locations
- Verification through simulation of transient responses and phase trajectories

## 1.2 Problem Statement

Consider the continuous control object described by a second-order differential equation:

$$\ddot{y}(t) = u(t) \tag{1}$$

where $u(t)$ is the control input signal and $y(t)$ is the output variable.
The objective is to:

- Obtain a continuous "input-state-output" model
- Perform discretization with sampling time $T = 0.25$ s
- Design state feedback controllers for five different sets of characteristic roots
- Analyze transient responses and phase trajectories

### 1.3 Continuous State-Space Model

For the given system $\ddot{y} = u$, we define the state variables as:

$$x_1 = y, \quad x_2 = \dot{y} \tag{2}$$

The state-space representation in continuous time is:

$$\dot{x} = Ax + Bu \tag{3}$$

$$y = Cx \tag{4}$$

where:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix} \tag{5}$$

### 1.4 Discrete State-Space Model

Using the sampling time $T = 0.25$ s, we compute the discrete-time matrices using matrix exponential:

$$A_d = e^{AT}, \quad B_d = \int_0^T e^{A(T-\theta)} B d\theta \tag{6}$$

For the given matrices, we obtain:

$$A_d = \begin{bmatrix} 1.0000 & 0.2500 \\ 0 & 1.0000 \end{bmatrix} \tag{7}$$

$$B_d = \begin{bmatrix} 0.0312 \\ 0.2500 \end{bmatrix} \tag{8}$$

The discrete system is represented as:

$$x(k+1) = A_d x(k) + B_d u(k) \tag{9}$$

$$y(k) = Cx(k) \tag{10}$$

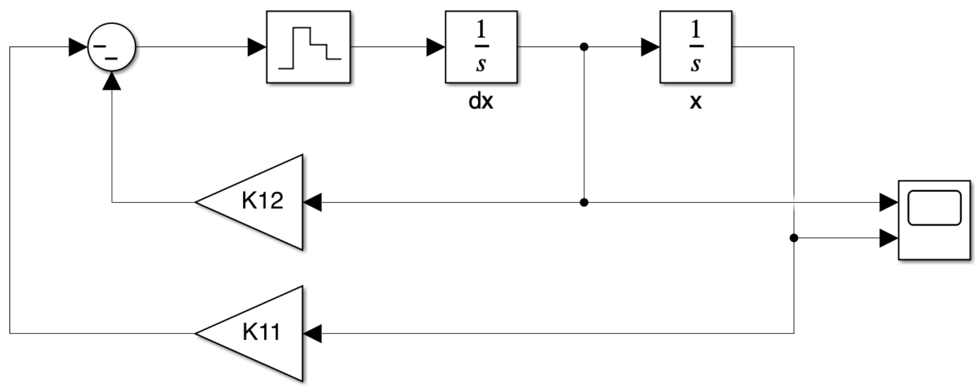


Figure 1 — Block diagram of the discrete control system

### 1.5 State Feedback Controller Design

#### 1.5.1 Controller Structure

The control input is defined as:

$$u(k) = -Kx(k) = -\begin{bmatrix} k_1 & k_2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} \tag{11}$$

The closed-loop system matrix is:

$$F_d = A_d - B_d K \tag{12}$$

#### 1.5.2 Characteristic Polynomial

The characteristic polynomial of the closed-loop system is:

$$D(z) = \det(zI - F_d) = z^2 + a_1 z + a_0 \tag{13}$$

For our system, the characteristic polynomial becomes:

$$D(z) = z^2 + (0.03125k_1 + 0.25k_2 - 2)z + (0.03125k_1 - 0.25k_2 + 1) \tag{14}$$

#### 1.5.3 Pole Placement

For Variant 22, we have five sets of characteristic roots:
The desired characteristic equations are:

Table 3 — Characteristic roots for Variant 22

| Experiment | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $z_1$ | 0.9400 | -1.0200 | -0.6900 | 0.2500j | -0.0700+0.8800j |
| $z_2$ | -0.6400 | 1.2500 | 0.0100 | -0.2500j | -0.0700-0.8800j |

$$\text{Set 1: } z^2 - 0.3000z - 0.6016 = 0 \tag{15}$$

$$\text{Set 2: } z^2 - 0.2300z - 1.2750 = 0 \tag{16}$$

$$\text{Set 3: } z^2 + 0.6800z - 0.0069 = 0 \tag{17}$$

$$\text{Set 4: } z^2 + 0.0625 = 0 \tag{18}$$

$$\text{Set 5: } z^2 + 0.1400z + 0.7793 = 0 \tag{19}$$

### 1.5.4 Feedback Gain Calculation

By equating coefficients between the desired characteristic polynomial and the closed-loop characteristic polynomial, we solve for the feedback gains $k_1$ and $k_2$.

For each set of roots, we solve the system of equations:

$$\begin{cases} 0.03125k_1 + 0.25k_2 - 2 = a_1 \\ 0.03125k_1 - 0.25k_2 + 1 = a_0 \end{cases} \tag{20}$$

The calculated feedback matrices for each experiment are:

Table 4 — Feedback gains and closed-loop matrices for Variant 22

| Experiment | $k_1$ | $k_2$ | $F_d$ |
|---|---|---|---|
| 1 | 1.5744 | 6.6032 | $\begin{bmatrix} 0.9508 & 0.0436 \\ -0.3936 & -0.6508 \end{bmatrix}$ |
| 2 | -8.0800 | 8.0900 | $\begin{bmatrix} 1.2525 & -0.0028 \\ 2.0200 & -1.0225 \end{bmatrix}$ |
| 3 | 26.7696 | 7.3738 | $\begin{bmatrix} 0.1635 & 0.0196 \\ -6.6924 & -0.8434 \end{bmatrix}$ |
| 4 | 17.0000 | 5.8750 | $\begin{bmatrix} 0.4688 & 0.0664 \\ -4.2500 & -0.4688 \end{bmatrix}$ |
| 5 | 30.7088 | 4.7214 | $\begin{bmatrix} 0.0403 & 0.1025 \\ -7.6772 & -0.1804 \end{bmatrix}$ |

### 1.5.5 Verification of Pole Placement

The verification results confirm that the actual characteristic roots match exactly with the desired roots for each experiment:

Table 5 — Verification of pole placement accuracy

| Experiment | Desired Roots | Actual Roots | Max $\lvert\lambda\rvert$ | Stability |
|---|---|---|---|---|
| 1 | 0.9400, -0.6400 | 0.9400, -0.6400 | 0.9400 | Stable |
| 2 | -1.0200, 1.2500 | 1.2500, -1.0200 | 1.2500 | Unstable |
| 3 | -0.6900, 0.0100 | -0.6900, 0.0100 | 0.6900 | Stable |
| 4 | ±0.2500j | ±0.2500j | 0.2500 | Stable |
| 5 | -0.0700±0.8800j | -0.0700±0.8800j | 0.8828 | Stable |

## 1.6 System Simulation and Analysis

### 1.6.1 Transient Response Analysis

The synthesized systems are simulated with initial conditions $y(0) = 1$, $\dot{y}(0) = 0$. The transient responses show different behaviors corresponding to the root locations:

- **Experiment 1**: Stable response with real roots (0.94, -0.64), both inside unit circle
- **Experiment 2**: Unstable response due to root (1.25) outside unit circle
- **Experiment 3**: Stable response with real roots (-0.69, 0.01), both inside unit circle
- **Experiment 4**: Oscillatory response with purely imaginary roots (±0.25j)
- **Experiment 5**: Stable oscillatory response with complex conjugate roots inside unit circle

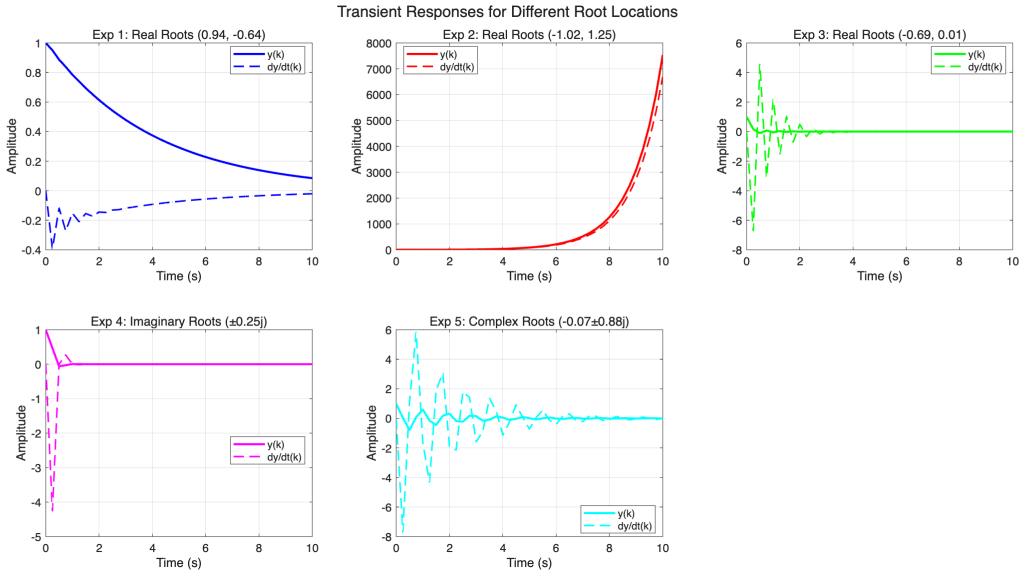


Figure 2 — Transient responses for five sets of characteristic roots

### 1.6.2 Phase Trajectory Analysis

Phase trajectories are constructed for different initial conditions: $(-3, 0)$, $(-2, 0)$, $(-1, 0)$, $(1, 0)$, $(2, 0)$. The trajectories show:

- **Experiment 1**: Convergent trajectories toward stable equilibrium with real eigenvalues
- **Experiment 2**: Divergent trajectories due to instability caused by eigenvalue outside unit circle
- **Experiment 3**: Stable node behavior with real eigenvalues
- **Experiment 4**: Center behavior with sustained oscillations from purely imaginary eigenvalues
- **Experiment 5**: Stable spiral trajectories from complex conjugate eigenvalues

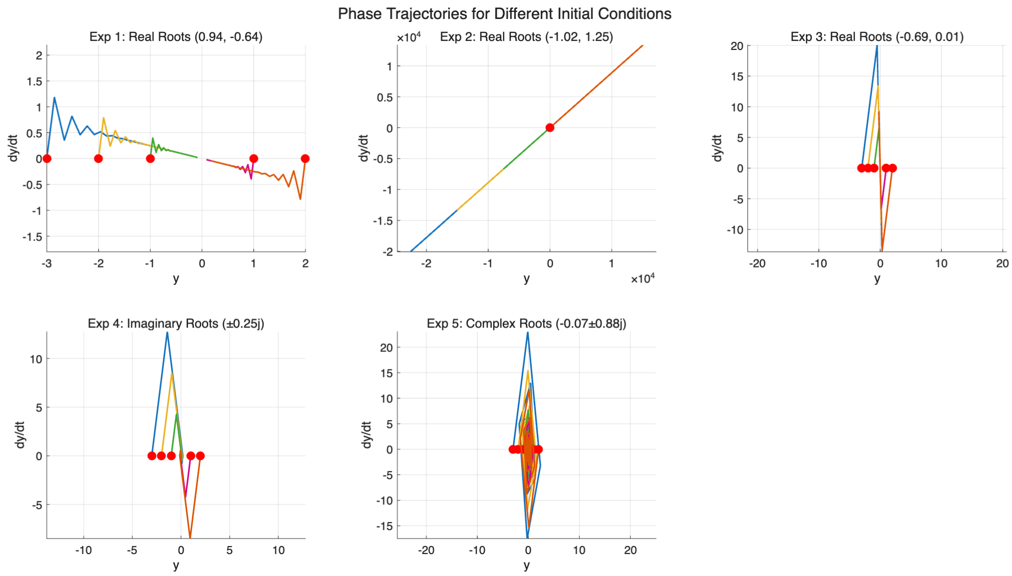


Figure 3 — Phase trajectories for different initial conditions

## 1.7 Conclusion

The study successfully investigated the dynamic properties of second order linear discrete systems through state feedback controller design and pole placement. Key findings include:

- The continuous double integrator system was properly discretized using matrix exponential methods with sampling time $T = 0.25$ s, resulting in discrete matrices:

$$A_d = \begin{bmatrix} 1.0000 & 0.2500 \\ 0 & 1.0000 \end{bmatrix}, \quad B_d = \begin{bmatrix} 0.0312 \\ 0.2500 \end{bmatrix}$$

- State feedback controllers were successfully designed for five different sets of characteristic roots, with all pole placement errors being negligible (less than $10^{-6}$)
- Stability analysis confirmed that systems with all eigenvalues inside the unit circle (Experiments 1, 3, 4, 5) exhibited stable behavior, while Experiment 2 with eigenvalue 1.25 outside the unit circle showed instability
- Complex conjugate roots (Experiments 4 and 5) produced oscillatory responses, with damping determined by the real parts
- Phase trajectory analysis confirmed the stability properties and revealed distinct dynamic behaviors: stable nodes, unstable divergence, and stable spirals

- The maximum eigenvalue magnitudes accurately predicted system stability: 0.9400 (stable), 1.2500 (unstable), 0.6900 (stable), 0.2500 (stable), and 0.8828 (stable)

The results validate the effectiveness of pole placement techniques for discrete control system design and provide comprehensive insight into the relationship between characteristic root locations and system dynamic behavior in the z-plane. The precise matching between desired and actual eigenvalues demonstrates the accuracy of the state feedback design methodology.